

1. Conditional Probability

Keywords to identify:

- “given that”
- “if it is known that”
- “provided that”
- “under the condition”

Concept:

One event has already occurred, and you are asked to find the probability of another event under that condition.

Mathematical Form:

$$P(A | B)$$

Formula:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Typical Example Pattern:

- “Probability that the second ball is green given that the first ball is red.”
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2. Bayes' Theorem

Keywords to identify:

- “it is found that”
- “given that the result is...”
- “find the probability that it came from...”
- “which bag/machine/source”

Concept:

You are given the result and asked to determine the source (reverse probability).

Direction:

Result → Cause

Mathematical Form:

$$P(\text{Cause} | \text{Result})$$

Formula:

$$P(A_i | B) = \frac{P(A_i) P(B | A_i)}{\sum P(A_j) P(B | A_j)}$$

Typical Example Pattern:

- “A ball is found to be red. Find the probability that it came from bag B.”
 - “A bolt is defective. Find the machine that produced it.”
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3. Binomial Distribution

Keywords to identify:

- “number of trials”
- “repeated experiments”
- “number of heads”
- “fires 6 times”
- “success/failure”
- “at most / at least / exactly”

Concept:

A fixed number of independent trials with two possible outcomes.

Structure:

$$X \sim \text{Binomial}(n, p)$$

Formula:

$$P(X = k) = {}^n C_k p^k q^{n-k}$$

Important Interpretations:

Phrase	Meaning
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At most k	$X \leq k$
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At least k	$X \geq k$
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Exactly k	$X = k$
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Typical Example Pattern:

- “Number of heads in 4 coin tosses”
- “Probability of hitting target in 6 attempts”

4. Discrete Random Variable (PMF)

Keywords to identify:

- “two dice”
- “minimum / maximum”
- “find distribution”
- “list possible values”
- “mean and variance”

Concept:

The random variable takes countable values, and probabilities are assigned to each value.

Steps to Solve:

1. List all possible values of X
2. Count favorable outcomes
3. Compute probabilities
4. Form the probability table (PMF)

Formulas:

$$E(X) = \sum xP(x)$$
$$Var(X) = E(X^2) - [E(X)]^2$$

5. PMF and CDF

Keywords to identify:

- “table of values”
- “find k ”
- “find $P(X \leq x)$ ”
- “distribution function”

Concept:

- PMF: Given probabilities
- CDF: Cumulative probability

Steps:

1. Use:

$$\sum P(x) = 1$$

to find unknown constant k

2. Compute CDF:

$$F(x) = P(X \leq x)$$

Method:

Add probabilities step by step up to the given value.

6. Continuous Random Variable (PDF)

Keywords to identify:

- “ $f(x) = \dots$ ”
- “integral”
- “check whether it is a PDF”
- “find mean and variance”
- “for $x \geq$ ” or “for interval”

Concept:

The variable takes values over a continuous range, and probabilities are found using integration.

PDF Conditions:

$$\begin{aligned} f(x) &\geq 0 \\ \int f(x) dx &= 1 \end{aligned}$$

Mean:

$$E(X) = \int xf(x) dx$$

Variance:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Probability over interval:

$$P(a < X < b) = \int_a^b f(x) dx$$

7. Core Identification Logic

Do not attempt to memorize entire solutions. Instead:

1. Read the question carefully
2. Identify key terms
3. Map them to the correct topic
4. Apply the corresponding formula

8. Master Identification Table

Keyword in Question	Topic
given that	Conditional Probability
result $\rightarrow$ find source	Bayes' Theorem
repeated trials	Binomial Distribution
at most / at least	Binomial Distribution
dice min/max	Discrete Random Variable
table of $P(x)$	PMF
$P(X \leq x)$	CDF
$f(x)$, integral	Continuous (PDF)

9. Final Strategy

- Focus on identifying the type of problem first
- Write the correct formula immediately
- Then proceed with calculations
- Maintain clean steps and proper limits in integrals